

YUANJUE (PERSY) ZHU

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EDUCATION

University of Michigan, Ann Arbor

Starting Fall 2026

Ph.D. in Mathematics

University of California, Berkeley

Sep, 2021 - Dec, 2025

B.A. in Mathematics with Honors

RELEVANT COURSEWORK

** indicates graduate level coursework; ** indicates supervised independent study*

Real Analysis; Honors Complex Analysis; Abstract Algebra; Abstract Linear Algebra; Probability Theory; Set Theory; Elementary Differential Topology; Differential Topology*; Lie Groups I*; Lie Groups II*; Introduction To Topology And Analysis II*; Riemannian Geometry*; Algebraic Topology*; Flow On Homogeneous Spaces*; Topology**; Quantum Theory, Groups and Representations**;
Dynamics, Geometry and Rigidity**

RESEARCH INTERESTS

Smooth Dynamics; Homogeneous Dynamics; Ergodic Theory; Riemannian Geometry; Geometric Group Theory

RESEARCH EXPERIENCE

Research Experience for Undergraduates (REU), University of Michigan

Summer 2025

Project supervised by Professor Ralf Spatzier

- Studied *Metric Spaces of Non-Positive Curvature* by Bridson–Haefliger and worked on classifying the visual boundaries of horospheres determined by different types of geodesics in CAT(0) space
- Produced a research report and presented the results during the REU talks; the report is available at this link

Undergraduate Researcher, Berkeley Artificial Intelligence Research

Fall 2023

Project supervised by Professor Aditi Krishnapriyan

- Worked on Weather and Latent SDE project using reinforcement learning agents under RLAX and Bsuite and implemented active-learning methods based on partial variational autoencoders

EMPLOYMENT EXPERIENCE

Undergraduate Reader, UC Berkeley

Fall 2024

Student Learning Center Math/Stat Tutor, UC Berkeley

Fall 2023

AWARDS, SCHOLARSHIPS AND PRESENTATIONS

Davis United World College Scholars Program

2021 - 2025

Joint Mathematics Meetings PME Contributed Section on Undergraduate Research (*Anticipated*)2026

SKILLS

Programming

L^AT_EX, Python, Java, SQL, JavaScript, Scheme, Pandas, Pytorch, Tensorflow, Jax, Git

Languages

English(Fluent); Chinese(Fluent); Spanish(Elementary Proficiency)